

Quantifying Fortune for Table Tennis Tournaments

Applied to the Paris 2024 Olympic Men's Singles Event

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Abstract

This paper develops a quantitative framework for measuring tournament draw favorability in professional table tennis. Using 390 clean matches from nine pre-Olympic tournaments, we estimate player strength via a three-feature model combining recent win rate, ITTF ranking score, and career achievement tier. Applied to the Paris 2024 Olympic bracket, the model produces a Draw Fortune Index measuring whether each player's bracket path was easier or harder than their seed predicts. Seeding explains only 17.3% of draw difficulty ($R^2 = 0.173$). Fortune correlates significantly with final placement (Pearson $r = -0.753$, $p < 0.001$; Spearman $r = -0.601$, $p < 0.001$). The luckiest player was Dang Qiu (Fortune = +0.752); the unluckiest was Truls Møregardh (Fortune = -2.081), who reached the final despite the hardest draw in the field.

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1 Introduction

In single-elimination tournaments, outcomes depend on player ability and on the sequence of opponents determined by the draw. A top seed may face elite opponents early while a lower seed advances through a weaker bracket.

The Paris 2024 Olympic table tennis men’s singles is a clear example. Wang Chuqin, the world’s top-ranked player (Score = 0.917), was eliminated in the second round. Fan Zhendong (Score = 0.737) won gold. This paper asks: did the draw, not skill, determine these outcomes?

Three questions guide the work:

1. Can a three-feature model reliably estimate player strength before the Olympics?
2. Can draw path difficulty be quantified into a Draw Fortune Index?
3. How accurately do predicted outcomes match actual results, round by round?

This work connects to tournament theory [2], the Bradley-Terry model [1], and mechanism design [3].

2 Theoretical Background

2.1 Tournament Theory and Mechanism Design

Tournament theory [2] shows that competition structure affects outcomes independently of ability. Mechanism design asks whether structures produce fair outcomes. The Draw Fortune Index tests whether Olympic seeding equalizes draw difficulty as intended.

2.2 The Bradley-Terry Model

Win probability for player i against j :

$$P(i \text{ beats } j) = \frac{\pi_i}{\pi_i + \pi_j}$$

Our three-feature score estimates π_i . All predicted match probabilities are computed from this formula.

2.3 Zero-Sum Games

Every match is zero-sum. The bracket is a sequence of such games where one loss ends a player’s tournament, amplifying the effect of any upset.

3 Data

3.1 Tournament Match Data

Match results were collected from nine international tournaments in the first half of 2024:

Table 1: Tournaments in dataset

Tournament	Matches
Singapore Smash 2024	63
Saudi Smash 2024	63
World Cup Macau 2024	15
WTT Champions Incheon 2024	31
WTT Champions Chongqing 2024	30
WTT Star Contender Doha 2024	47
WTT Star Contender Goa 2024	47
WTT Star Contender Ljubljana 2024	47
WTT Star Contender Bangkok 2024	47
Total (after cleaning)	390

Two additional datasets support the strength model: ITTF World Rankings (Week 30, July 23 2024) and a manually constructed career achievement tier table (scores 1 to 7). The Paris 2024 Olympic bracket was sourced from Wikipedia.

3.2 Data Cleaning

All names standardized to uppercase. BYE rows removed (390 matches remain). WinnerName derived from the Winner (1 or 2) column. A separate per-player win-rate table was built from these 390 matches; players with fewer than three matches were excluded from that table only, so opponents’ records in the underlying match data remain intact. Six name mismatches across datasets (e.g., *Jang Woo-Jin* vs. *Jang Woojin*, *Wang Yang* vs. *Yang Wang*) were resolved manually. Players absent from tournament data received median feature values. Final dataset: 390 matches, 74 players with win rate data, 49 Olympic players modeled.

One additional data-quality note: 16 of the 19 players eliminated in the Round of 32 had no recorded opponent in the source bracket file. For these players, Draw Difficulty was imputed using the median player Score rather than a specific opponent’s score; their Fortune values are therefore approximate. See the limitations discussion in Section 8.

4 Three-Feature Strength Model

4.1 Feature 1: Recent Win Rate

$$\text{WinRate}_i = \frac{\text{Wins}_i}{\text{Matches}_i}$$

Captures current form from the nine pre-Olympic tournaments.

4.2 Feature 2: ITTF Ranking Score

$$\text{RankScore}_i = \frac{\text{Points}_i}{\max_j(\text{Points}_j)}$$

Wang Chuqin held 7925 points (RankScore = 1.000). Captures long-run accumulated performance across all official competitions.

4.3 Feature 3: Career Achievement Score

Each player is assigned a tier score for their best career result before Paris 2024, normalized by dividing by 7:

Table 2: Career achievement tiers

Tier	Achievement	Example
7	Olympic Champion	Fan Zhendong
6	World Champion / Olympic Silver	Wang Chuqin, Möregardh
5	Olympic Bronze	Félix Lebrun, Ovtcharov
4	Olympic 4th / Team Gold	Calderano, Jang Woo-Jin
3	World SF / European Champion	Harimoto, Dang Qiu
2	World QF / European medal	Mid-tier players
1	National level only	Lower seeds

Normalized: $\text{AchievScore}_i = \text{Tier}_i / 7$.

4.4 Combined Score

The three features are combined with equal weights:

$$\text{Score}_i = \frac{1}{3} (\text{WinRate}_i + \text{RankScore}_i + \text{AchievScore}_i)$$

Table 3: Top 15 players by strength score

Rank	Player	WinRate	RankScore	AchievScore	Score
1	Wang Chuqin	0.895	1.000	0.857	0.917
2	Fan Zhendong	0.750	0.462	1.000	0.737
3	Félix Lebrun	0.688	0.453	0.714	0.618
4	Hugo Calderano	0.643	0.447	0.571	0.554
5	Dimitrij Ovtcharov	0.556	0.194	0.714	0.488
6	Tomokazu Harimoto	0.667	0.332	0.429	0.476
7	Dang Qiu	0.692	0.254	0.429	0.458
8	Jang Woo-Jin	0.583	0.210	0.571	0.455
9	Lin Yun-Ju	0.583	0.336	0.429	0.449
10	Cho Dae-Seong	0.625	0.150	0.571	0.449
11	Truls Möregardh	0.333	0.127	0.857	0.439
12	Darko Jorgic	0.444	0.153	0.429	0.342
13	Sharath Kamal Achanta	0.600	0.081	0.286	0.322
14	Ovidiu Ionescu	0.600	0.073	0.286	0.319
15	Omar Assar	0.375	0.135	0.429	0.313

Truls Möregardh (Rank 11) illustrates the model’s value: his low WinRate (0.333) and RankScore (0.127) are offset by a high AchievScore (0.857), giving a more balanced estimate than win rate alone.

5 Draw Difficulty and Fortune Index

5.1 Draw Difficulty

$$\text{Difficulty}_i = \sum_{r \in \text{Rounds}} \text{Score}_{\text{opp}(i,r)}$$

For each round a player competed in, the Score (Section 4) of their actual opponent is added. BYEs are excluded. A higher total means the player walked a harder bracket path. For the 16 players whose Round of 32 opponent was not recorded, the median player Score is substituted for that round's contribution.

5.2 Expected Difficulty

Difficulty alone cannot say whether a draw was unfair, since top seeds are designed to face stronger opponents than low seeds. To isolate the unfair component, we estimate how much Difficulty a player's seed alone would predict using ordinary least squares (OLS) linear regression:

$$\text{ExpectedDifficulty}_i = \hat{\alpha} + \hat{\beta} \cdot \text{Seed}_i$$

where the coefficients are obtained by minimizing the sum of squared residuals across all 49 Olympic players:

$$(\hat{\alpha}, \hat{\beta}) = \arg \min_{\alpha, \beta} \sum_{i=1}^{49} (\text{Difficulty}_i - \alpha - \beta \cdot \text{Seed}_i)^2$$

This produces $\hat{\alpha} = 0.981$ and $\hat{\beta} = -0.0158$ ($R^2 = 0.173$, $p < 0.01$). The negative slope confirms that lower seed numbers (stronger players) faced higher Difficulty on average, consistent with how Olympic seeding is designed to work. However, seeding explains only 17.3% of the variation in Difficulty; the remaining 82.7% reflects structural randomness in the bracket draw.

Worked example. For seed 7 (Dang Qiu):

$$\text{ExpectedDifficulty} = 0.981 - 0.0158 \times 7 = 0.870 \approx 0.871$$

For seed 19 (Möregardh):

$$\text{ExpectedDifficulty} = 0.981 - 0.0158 \times 19 = 0.681$$

5.3 Draw Fortune Index

$$\text{Fortune}_i = \text{ExpectedDifficulty}_i - \text{Difficulty}_i$$

Positive Fortune means the player's actual path was easier than their seed predicted (lucky); negative Fortune means it was harder (unlucky).

6 Results

6.1 Draw Fortune Index

Table 4: Most and least fortunate players, Paris 2024

Seed	Player	Difficulty	Exp. Diff.	Fortune	Result
<i>Most fortunate (easiest draw relative to seed)</i>					
7	Dang Qiu	0.118	0.871	+0.752	2nd round
12	Marcos Freitas	0.221	0.791	+0.571	1st round [†]
14	Quadri Aruna	0.221	0.760	+0.539	1st round [†]
15	Cho Dae-Seong	0.221	0.744	+0.523	1st round [†]
22	Edward Ly	0.221	0.633	+0.412	1st round [†]
<i>Least fortunate (hardest draw relative to seed)</i>					
19	Truls Möregardh	2.762	0.681	−2.081	Silver
3	Félix Lebrun	2.462	0.934	−1.528	Bronze
59	Kanak Jha	1.179	0.048	−1.131	Round of 16
2	Fan Zhendong	1.705	0.950	−0.755	Champion
39	Wang Yang	0.917	0.364	−0.553	1st round

[†]Fortune values for these players are approximate: their Round of 32 opponent was not recorded in the source file and was substituted with the median player Score. Dang Qiu’s result is not affected by this imputation.

Fortune correlates significantly with final result (Pearson $r = -0.753$, $p < 0.001$; Spearman $r = -0.601$, $p < 0.001$). Players with harder draws tended to advance further, reflecting the seeding system’s design: stronger players are placed into harder bracket halves by construction.

6.2 Round-by-Round Accuracy

The bracket begins with a Preliminary round (seeds 65 to 67 only), followed by the Round of 32, Round of 16, Quarterfinal, Semifinal, and Final, with a separate Bronze medal match. Table 5 reports prediction accuracy at each stage using the three-feature Score and the Bradley-Terry formula.

Table 5: Prediction accuracy by round

Round	Matches	Correct	Accuracy	Upsets
Preliminary	2	1	50.0%	1
Round of 32	19	15	78.9%	4
Round of 16	8	8	100.0%	0
Quarterfinal	4	4	100.0%	0
Semifinal	2	1	50.0%	1
Final and Bronze match	2	2	100.0%	0
Overall	37	31	83.8%	6

The model is perfect from the Round of 16 onward except for one semifinal upset, where score differentials are largest. All six upsets occur in the Preliminary round, Round of 32, or Semifinal.

6.2.1 Round of 32 Upsets

Table 6: Round of 32 upsets

Winner	Defeated	Winner Score	Loser Score	$\hat{P}$ (loser wins)
Truls Möregardh	Wang Chuqin	0.404	0.917	0.306
Kirill Gerassimenko	Dang Qiu	0.118	0.458	0.205
Anders Lind	Milosz Redzinski	0.113	0.060	0.655
Kanak Jha	Panagiotis Gionis	0.055	0.214	0.204

Three upsets had predicted probabilities below 0.22. The Anders Lind vs. Milosz Redzinski match is the exception: both players had low, similar scores and the outcome is consistent with near-random results at small score differentials.

6.2.2 Round of 16 and Quarterfinal

All twelve matches in these two rounds were correctly predicted. Selected results:

Table 7: Round of 16 and Quarterfinal, selected matches

Winner	Defeated	Winner Score	Loser Score	$\hat{P}$ (winner)
Fan Zhendong	Kanak Jha	0.737	0.055	0.931
Tomokazu Harimoto	Anders Lind	0.476	0.113	0.809
Hugo Calderano	Alexis Lebrun	0.554	0.259	0.681
Fan Zhendong	Tomokazu Harimoto	0.737	0.476	0.608
Truls Möregardh	Omar Assar	0.404	0.313	0.563

6.2.3 Semifinal, Final, and Bronze Match

Table 8: Semifinal, Final, and Bronze medal match

Stage	Winner	Defeated	Winner Score	Loser Score	$\hat{P}$ (winner)
Semifinal	Fan Zhendong	Félix Lebrun	0.737	0.618	0.544
Semifinal	Truls Möregardh	Hugo Calderano	0.404	0.554	0.422*
Bronze	Félix Lebrun	Hugo Calderano	0.618	0.554	0.527
Final	Fan Zhendong	Truls Möregardh	0.737	0.404	0.646

*Probability shown is $\hat{P}$ (Calderano wins = 0.578); Möregardh’s win probability was $1 - 0.578 = 0.422$, making this the sole semifinal upset. The model assigned Calderano a 57.8% chance of victory, but Möregardh prevailed. This result illustrates that substantial uncertainty remains when player strength scores are close.

6.3 Key Match Analysis

Wang Chuqin vs. Truls Möregardh (Round of 32). The tournament’s defining upset. Wang (Score = 0.917) was the heavy favorite by the three-feature model but lost to Möregardh (Score = 0.439), who was given only a 30.6% chance of winning. Möregardh (Fortune = -2.081) then

advanced through the hardest recorded draw in the field, defeating Kao Cheng-Jui, Omar Assar, and Hugo Calderano before losing to eventual champion Fan Zhendong in the final.

Dang Qiu vs. Kirill Gerassimenko (Round of 32). Dang Qiu (Score = 0.458, Fortune = +0.752) lost in the second round despite having the easiest draw, by Fortune, of any confirmed player in the field. His result illustrates the paper’s key finding: a favorable draw is an opportunity, not a guarantee of advancement.

7 Discussion

Main findings:

1. Seeding explains only 17.3% of draw difficulty ($R^2 = 0.173$). Over 80% of variation is structural randomness, showing the draw mechanism introduces meaningful inequality beyond seeding rank.
2. Fortune is a significant predictor of results (Pearson $r = -0.753$, $p < 0.001$), confirming that bracket path plays a measurable role. However, this correlation partly reflects the seeding system’s own design: stronger players are deliberately placed in harder bracket halves, so higher Fortune and higher result scores are both downstream of player strength.
3. Möregardh’s run (Fortune = -2.081) is the tournament’s defining anomaly: the hardest structural path combined with a run to the final. Wang Chuqin’s elimination opened the upper bracket half, meaning Möregardh avoided another meeting with the top seed, but he still had to defeat three top-eight players to reach the final.
4. Dang Qiu’s exit (Fortune = +0.752) shows that the easiest confirmed draw in the field did not translate into a deep run, reinforcing that Fortune measures structural opportunity, not outcome.

Limitations. The Round of 32 opponent was unrecorded for 16 of 19 players eliminated in that round; their Difficulty was imputed using the median player Score, so their individual Fortune values carry more uncertainty than players with fully recorded paths. Nine of the 49 Olympic players had no win-rate data from the nine pre-Olympic tournaments and received the median WinRate. World Cup Macau, WTT Champions Incheon, and WTT Champions Chongqing yielded fewer matches in the available source files than the original tournament totals would suggest, slightly limiting win-rate precision for players active mainly in those events. Equal feature weights were not optimized. The cumulative Difficulty metric (summing opponents across all rounds survived) may partially conflate draw difficulty with tournament survival, since players who advance further accumulate more terms regardless of draw quality; a fully ex-ante measure restricted to pre-determined opponents would address this, and is identified as the primary direction for future work.

Future work. Reconstruct the full bracket tree from official ITTF records to compute a genuinely ex-ante difficulty measure at every round, independent of realized results. Optimize feature weights via logistic regression. Extend to women’s singles and prior Olympic cycles.

8 Conclusion

A three-feature player strength model and a Draw Fortune Index were constructed for the Paris 2024 Olympic table tennis men’s singles using a fully reproducible data pipeline. The model achieves 83.8% match prediction accuracy (31 of 37). Seeding explains only 17.3% of draw difficulty, and Fortune correlates significantly with final placement (Pearson $r = -0.753$, $p < 0.001$). Truls Möregårdh, the most structurally disadvantaged player by this measure, produced the tournament’s most remarkable performance by advancing to the final from the hardest recorded bracket path in the field, while Dang Qiu’s early exit despite the easiest confirmed path underscores that a favorable draw is an opportunity, not a guarantee.

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